

EE5141 Wireless and Cellular Communication

Simulation Assignment 2

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1 Frequency Synchronization

The preamble uses a Schmidl-Cox structure with QPSK symbols placed at odd subcarriers, creating a symmetric time-domain waveform with an $N/2$ -spaced repetition.

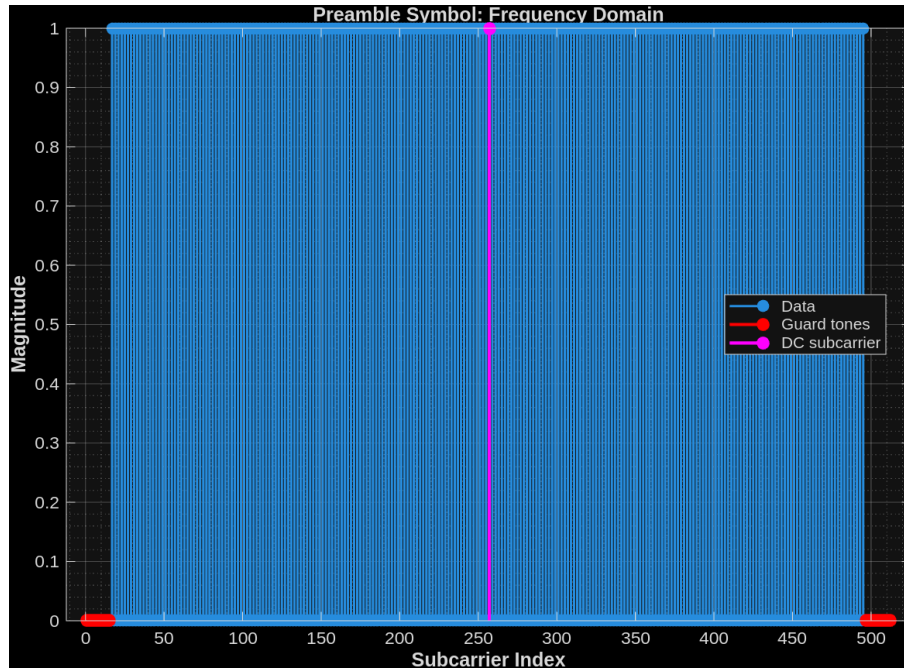


Figure 1: Preamble Subcarrier Structure

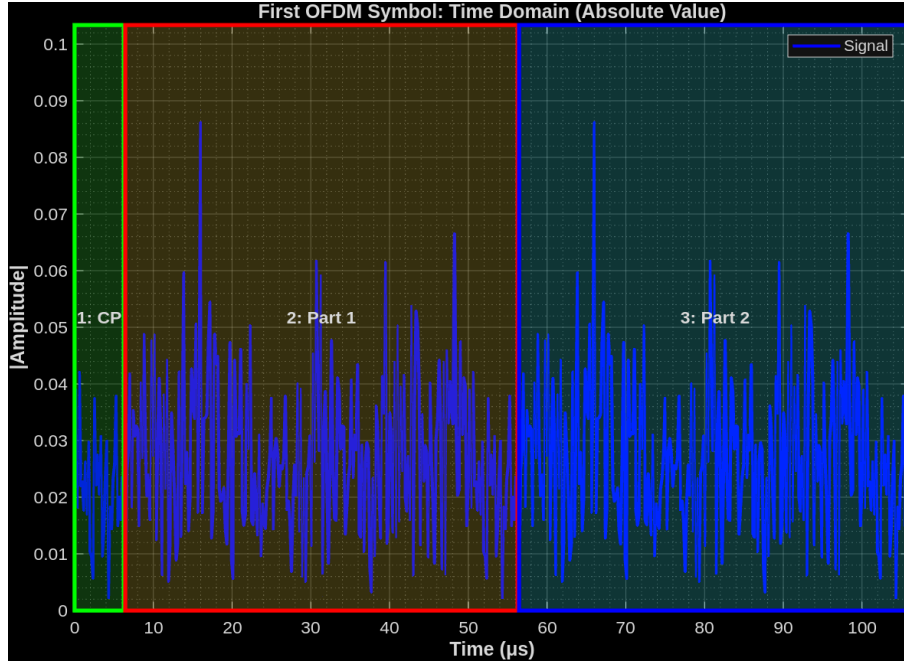


Figure 2: Time-Domain Preamble Waveform

Key Findings

- The maximum unambiguous frequency offset that can be detected equals half the subcarrier spacing:

$$CFO_{\max} = \frac{\Delta f}{2} = 5 \text{ kHz}$$

- This limitation arises because the autocorrelation is computed over $N/2$ samples.
- If the CFO exceeds this limit, aliasing and estimation ambiguity occur.
- The provided CFO of 28.65 kHz exceeds this limit and therefore cannot be reliably estimated.
- The MSE versus SNR performance is therefore evaluated for a reduced CFO of 2.865 kHz, which lies within the unambiguous estimation range.

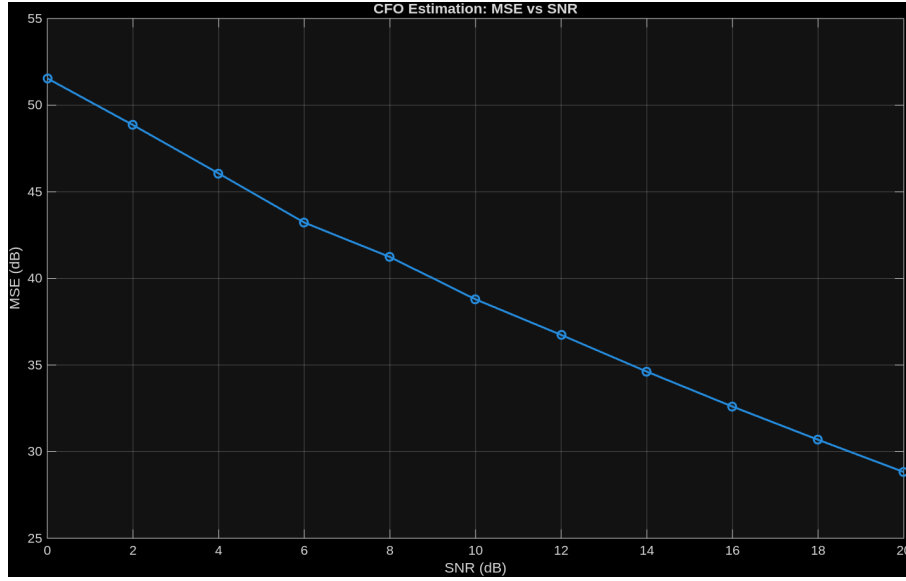


Figure 3: CFO Estimation MSE versus SNR

2 Timing Synchronization

The repeating $N/2$ -spaced structure in the Schmidl-Cox preamble is exploited to estimate timing offset through autocorrelation of the received signal.

For two OFDM frames with padded zeros, the normalized autocorrelation under different channel conditions (AWGN-only and multipath at $\text{SNR} = 6$ dB) is shown below.

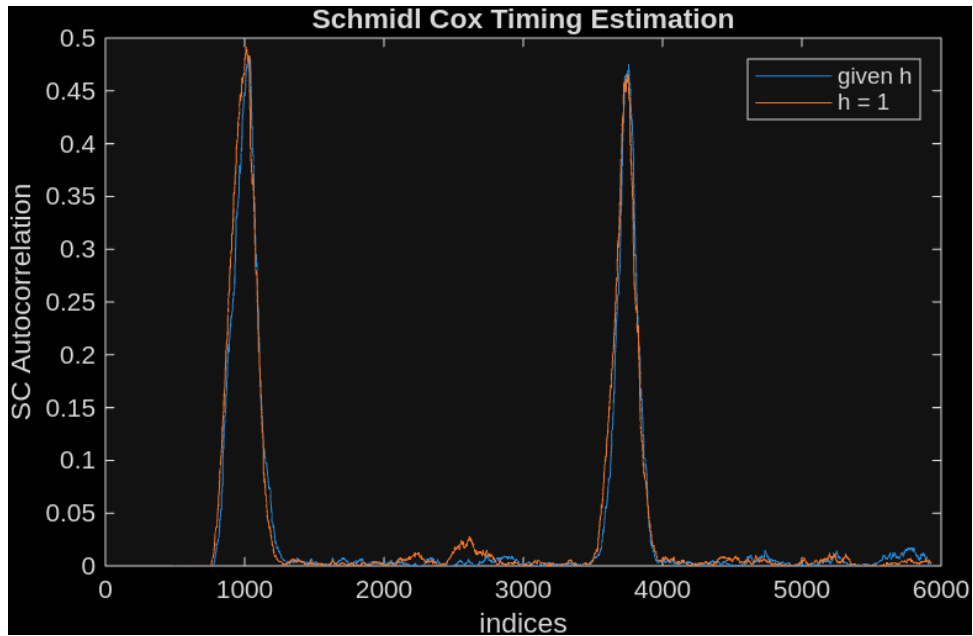


Figure 4: Normalized Autocorrelation for Two OFDM Frames

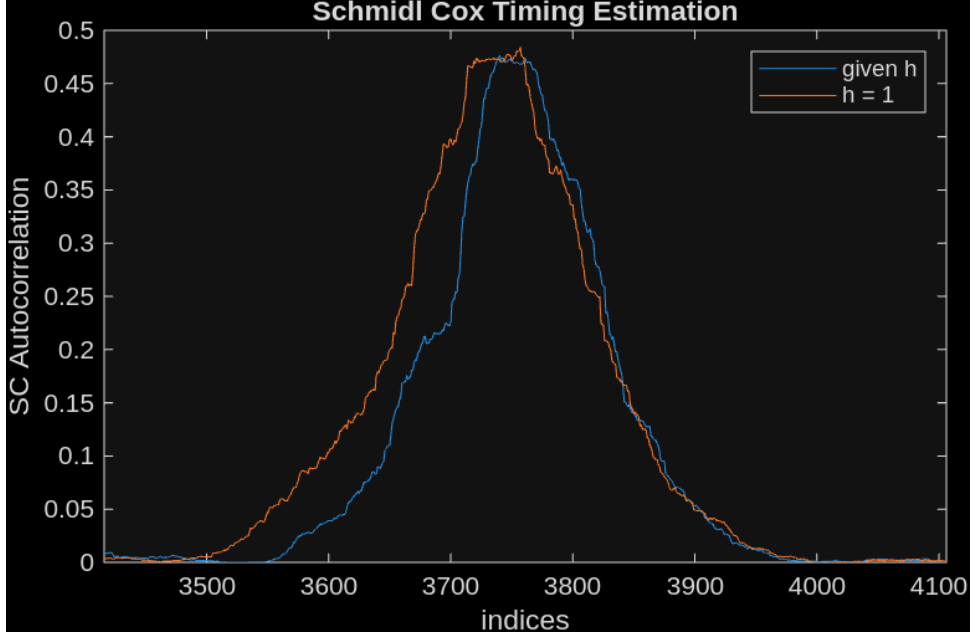


Figure 5: Zoomed View of the Timing Metric Plateau

Observations

- A characteristic plateau appears for each OFDM frame due to the repetition structure in the preamble.
- The AWGN-only channel exhibits a wider plateau compared to the multipath channel.
- The plateau width is given by

$$\text{Width} = L_{\text{cp}} - \tau_{\text{max}} + 1$$

where τ_{max} denotes the maximum channel delay.

3 Channel Estimation

The MSE versus SNR performance for five different channel estimation methods is compared below.

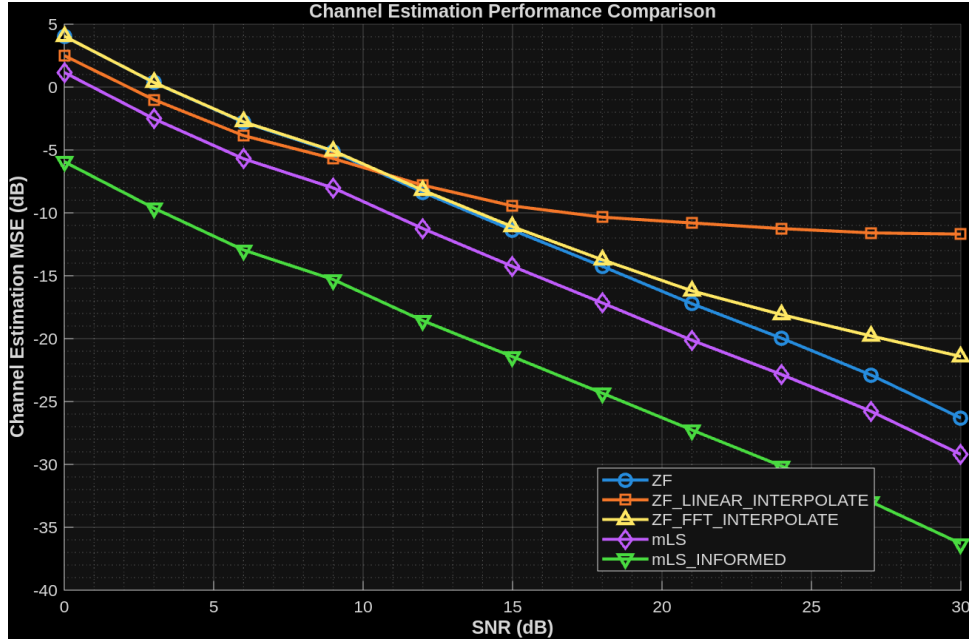


Figure 6: Channel Estimation MSE versus SNR

Observations

1. Zero-Forcing (ZF) Performance

- Basic ZF estimation is performed only at pilot subcarriers; interpolation is required to recover the channel over all subcarriers.
- FFT-based sinc interpolation significantly outperforms linear interpolation because it better exploits the frequency-domain structure of the OFDM signal.
- Since the MSE is evaluated only at pilot indices, the basic ZF estimator appears to achieve artificially low MSE.

2. Maximum Likelihood Sequence (MLS) Performance

- Standard MLS estimation outperforms ZF interpolation methods by utilizing an optimal least-squares framework.
- The informed MLS estimator (`mLS_informed`) assumes prior knowledge of the channel tap delays, resulting in superior estimation accuracy.

3. Relative Performance Ranking

`mLS_informed` > `mLS` > ZF (FFT interpolation) > ZF (linear interpolation) > ZF